7 Two blocks of masses m and $2m$ approach each other with speeds u and $0.5u$ and collide head-on. A light spring of force constant k is attached to the block of mass $2m$.

What is the maximum compression experienced by the spring during the collision?

A $\sqrt{\frac{mu^2}{6k}}$ **B** $\sqrt{\frac{2mu^2}{3k}}$ **C** $\sqrt{\frac{3mu^2}{2k}}$ **D** $\sqrt{\frac{3mu^2}{k}}$